

3.3 Integrated Development Environment (IDE) & VS Code

آئی ڈی ای (IDE) ایک ایسا سافٹ ویئر ہے جس میں کوڈ لکھنا، چلانا اور غلطیاں درست کرنا ایک ہی جگہ ممکن ہوتا ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 03 of 13

3.3 Integrated Development Environment (IDE) & VS Code

LEARN IT

An **IDE (Integrated Development Environment)** is a software tool that helps programmers **write, run and manage code in one place**. It is a complete workspace for programming. "Integrated" means many tools are joined together in one window.

Key features of an IDE:

- **Code Editor:** where you write and edit code.
- **Syntax Highlighting:** colours different parts of the code (keywords, strings, numbers) to make it easier to read.
- **Auto-completion:** suggests code while you type (type `pri` and it offers `print`).
- **Run/Execute Tool:** runs the program with one click.
- **Debugger:** helps you find and fix errors, for example by pausing the program at a chosen line.

WORKED EXAMPLE

Without vs with an IDE. Without an IDE you would write code in Notepad, save it, open a terminal and type `python file.py`, then go back to Notepad to fix mistakes. With an IDE you write, run (press the ► Run button) and debug in the same window, and errors are underlined while you type.

REMEMBER IT

IDE features — "**E-S-A-R-D**": Editor, Syntax highlighting, Auto-completion, Run, Debugger.

COMMON MISTAKE

An IDE does not write the program for you and does not catch logical mistakes. It only makes writing, running and debugging easier.

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CHECK YOURSELF

1. What does "integrated" mean in IDE?
2. Which IDE feature suggests code while you type?

Answers: 1. Many programming tools (editor, runner, debugger) are combined in one program. 2. Auto-completion.

LEARN IT

VS Code is a **free, lightweight and powerful code editor** made by Microsoft and widely used for Python. By default it is **not a full IDE**, but it becomes one when you install **extensions** (add-ons), such as the **Python extension by Microsoft**.

VS Code interface elements

Element	Description
1. Top Menu Bar	Control over files, editing, navigation and workspace (File, Edit, Selection, View, Go, Run).
2. Side Bar	Contains the core tools: Explorer, Search, Source Control, Run & Debug and Extensions.
3. Explorer	Shows project files and folders. Helps you create, open and manage your Python files.
4. Search	Finds and replaces text in the open file or across the whole project.
5. Source Control	Connects with Git (a version control system) to track changes, undo mistakes and collaborate.
6. Run & Debug	Runs code, finds errors, pauses at specific lines and lets you inspect variables.
7. Extensions	Like an App Store for VS Code. Add features such as the Python extension, Prettier or Python Debugger.



A simplified VS Code window: menu bar, side bar, Explorer and the code editor.

WORKED EXAMPLE

Setting up and running your first program in VS Code:

Install Python → Install VS Code → Extensions: install "Python" (Microsoft) →
 Explorer: new file hello.py → Type code → Click ▶ Run

```
print("Hello, World!")
```

OUTPUT

```
Hello, World!
```

The output appears in the **Terminal** panel at the bottom of the window.

REMEMBER IT

Extensions are the **"App Store"** of VS Code: just as apps give your phone new abilities, extensions give VS Code new abilities.

COMMON MISTAKE

Always save Python files with the `.py` extension (e.g. `hello.py`). Without it, VS Code does not know the file is Python, so there is no syntax highlighting and no Run button.

اردو میں سمجھیں

وی ایس کوڈ (VS Code) ایک مفت اور ہلکا پھلکا کوڈ ایڈیٹر ہے۔ یہ اپنے آپ میں مکمل آئی ڈی ای نہیں، لیکن ایکسٹینشنز (Extensions) لگانے سے آئی ڈی ای بن جاتا ہے۔ اس کے اہم حصے مینو بار، سائیڈ بار، ایکسپلورر، سرچ، سورس کنٹرول، رن اینڈ ڈیپگ اور ایکسٹینشنز ہیں۔

CHECK YOURSELF

1. Which VS Code element connects to Git?
2. Is VS Code a full IDE by default? How does it become one?
3. Where do you see and manage your project's files?

Answers: 1. Source Control. 2. No; it becomes an IDE by installing extensions (e.g. the Python extension). 3. Explorer.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An IDE is:

- a. One app to write, run and debug code
- b. A printer
- c. A sort
- d. A cable

MCQ 2. VS Code is:

- a. A popular editor/IDE
- b. A DBMS
- c. An OSI layer
- d. A gate

MCQ 3. The terminal/run button is for:

- a. Executing the program
- b. Drawing pie charts only
- c. Sending TCP SYN only
- d. K-maps

MCQ 4. Syntax highlighting helps you:

- a. See keywords and mistakes
- b. Sort faster
- c. Encrypt disks
- d. Route packets

MCQ 5. A file explorer in VS Code shows:

- a. Project files
- b. OSI layer 1
- c. NAND groups
- d. Beachhead users

MCQ 6. Debugging in an IDE uses:

- a. Breakpoints and variable watch
- b. Only bubble sort
- c. Only pie charts
- d. Only Agile stickers

Extra short questions

1. Define IDE.
2. Name four VS Code areas.
3. Why not use Notepad only?
4. What is an extension?

Extra long question

1. Describe how you would create, run and debug a first Python file in VS Code. Include two interface elements from the notes.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Integrated Development Environment: editor + run + debug + files in one place.

2. Activity bar/sidebar (files), editor, terminal, status bar (and extensions).
3. No easy run, no highlighting, no debugger.
4. An add-on (Python plugin) that teaches VS Code a language.

Long

1. New file .py, type print, Run Python, read terminal. Mention top menu, explorer, terminal panel. Breakpoint on a wrong variable.

Next lecture: 3.4 Fundamentals of Python Programming